

Deterministic Optimization

SOC, SDP Examples

Andy Sun

Assistant Professor

Stewart School of Industrial and Systems Engineering

Statistical Classification Problem

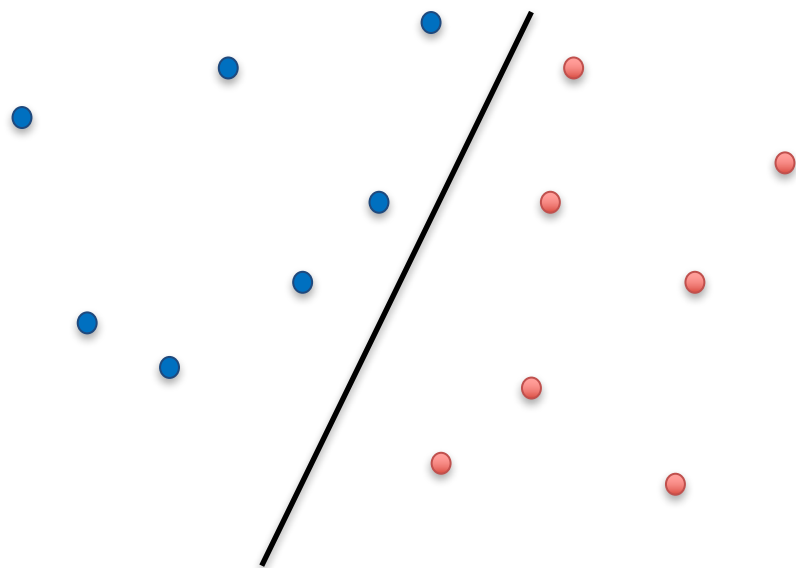
SOCP, SDP Examples

Learning Objectives for this module

- Discover important applications of SOCP and SDP for statistical classification, experiment design, and geometric extremal problems
- For this lesson, we focus on statistical classification problem

Linear Classification

- Consider two sets of points $\{x_1, \dots, x_N\}$ (blue) and $\{y_1, \dots, y_M\}$ (red)
- Want to find an affine function to classify blue and red points

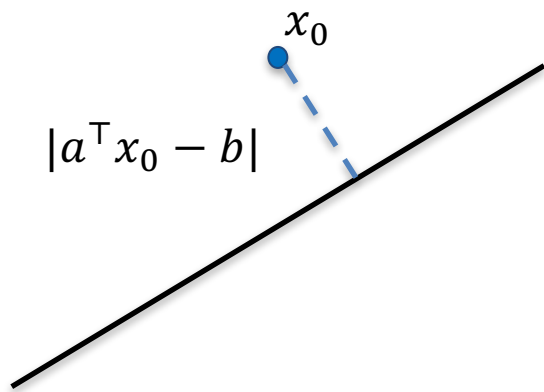


$$a^\top x_i - b > 0 \quad \forall i = 1, \dots, N$$
$$a^\top y_i - b < 0 \quad \forall i = 1, \dots, M$$

Measure Distance to Hyperplane

- Given a point x_0 and a hyperplane defined as $a^\top x - b = 0$ with $\|a\|_2 = 1$
- The Euclidean distance from x_0 to the hyperplane is given as

$$|a^\top x_0 - b|$$



Robust Linear Classification

- We want to find the affine function that gives the maximum separation of the two sets of point

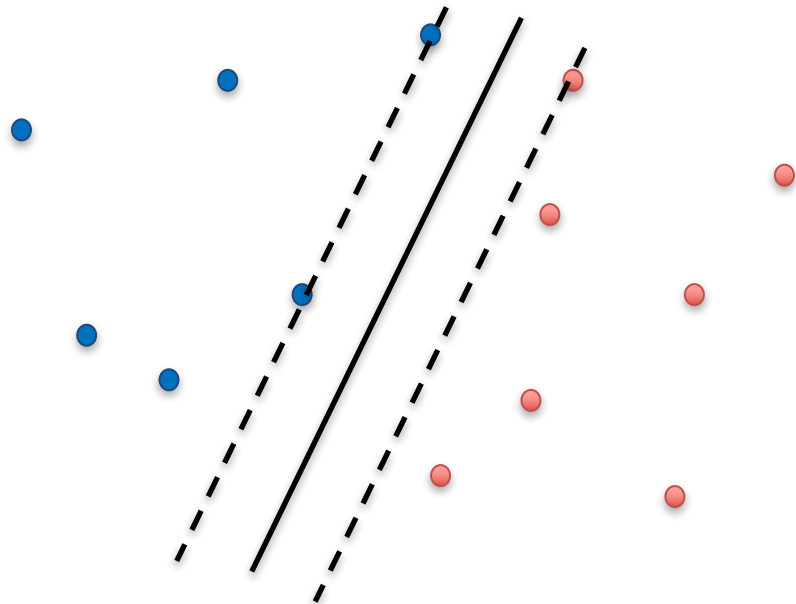
- $\max_{a,b,t} t$

s.t. $a^\top x_i - b \geq t \quad \forall i = 1, \dots, N$

$$a^\top y_i - b \leq -t \quad \forall i = 1, \dots, M$$

$\|a\|_2 \leq 1$

SOCP constraint



Summary

- We learned:
 - Basic robust statistical classification problem
 - How to formulate it as an SOCP